Input Data and Parameters

Forecast appliance operation (24 h)

- Activation profiles
- Power modulation profiles

Device parameters

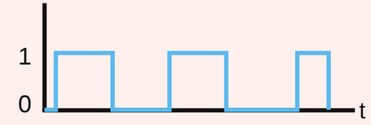
- Flexibility type
- Power Max
- Weight bounds
- Modulation bounds
- Operating constraints

Kitchen parameters

- Power and energy limits
- Electricity tariffs
- Operational timezones

1 Data-Driven Device Operational Modelling

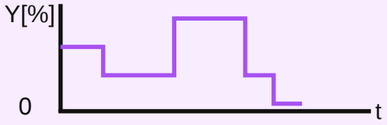
Time-Flexible Devices



Characterization

- Minute-level characterization (t)
- Block-level characterization (k)

Power-Flexible Devices



Characterization

- Minute-level characterization (t)
- Block-level characterization (k)

Refrigeration Temperature Reconstruction

(time-flexible devices)

2 Day-Ahead Planning (24-hour horizon)

Input:

- Forecast appliance operation
- Device parameters
- Kitchen parameters
- Block-level characterization (step 1)

Day-ahead optimization (MILP)

- Utility / Energy / Cost objective
- Device-level constraints
- Kitchen-level constraints

Day-ahead plan

- Scheduled device operation (per block)



B-min blocks

- Flexibility weight references (time- and power-flexible devices)

3 Real-Time Control (1-minute resolution)

Input:

- Forecast appliance operation
- Device parameters
- Kitchen parameters
- Minute-level characterization (step 1)

Feasibility checks

- Time flexibility bounds (force ON/OFF)
- Power-hold constraints

Real-time optimization (MILP)

- Utility / Energy / Tracking objective
- Device operating constraints
- Aggregate power limit (per minute)

Implemented operation (1-min)

- Device operating states
- Updated flexibility weights



Outputs and Performance Evaluation

Appliance operation
Activation states
Power modulation levels
Weight behaviour

Aggregate load profile
24-hour power and energy consumption

Energy cost
Total cost
Cost breakdown

Limit violations
Power/energy compliance
Violation metrics

Refrigerator thermal response
Reconstructed temperature

Computational performance
Optimisation times and gap